Can AI-generated data validation rules be trusted alongside expert controls?

MSc Artificial Intelligence thesis proposal

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1 Problem and research objective

Data teams check incoming data before it is used in reports, analyses or other systems. These checks range from simple technical controls, such as required fields and valid codes, to business rules that relate several fields to each other. At Rabobank, this work includes technical validation, business rules, monitoring thresholds and the manual investigation of exceptions [1]. Banks must also be able to show that such controls exist and work: the Basel Committee's principles for risk data (BCBS 239) and the European Central Bank's 2024 guide treat data quality and its controls as part of sound risk management [2, 3].

Most of these controls are validation rules. A validation rule is a condition that every record must meet, for example "total assets must equal total liabilities plus equity" in a balance sheet. When a record does not meet the condition, the rule raises an alert, and a person investigates whether the data is wrong. Writing good rules requires knowledge of the data and of the business, and maintaining them takes time.

A language model can write candidate validation rules from a description of the data within seconds, and data-quality tools have started to offer this [4]. Some of these rules are useful. Others apply a wrong business assumption, repeat a control that already exists, or raise alerts on records that are correct. A rule such as "the loan amount must be below five times the income" may look reasonable, but it is not a valid data-quality rule unless the business has such a policy. In addition, the same model can write different code when asked the same question twice [5], and rules that are discovered automatically from data often turn out to be false [6].

For a team that already maintains many checks, three questions therefore matter. Does a generated rule do what it is supposed to do? Does it catch errors that the existing controls miss, and how many extra alerts does that cost? And would the rule have behaved well on past data, so that the team can decide whether it may stop data automatically, may only warn, or should only be monitored?

This thesis studies these three questions. It measures how correctly and consistently language models implement documented expert rules, how much they add beyond independent expert controls, which method best selects a small set of useful rules, and how the selected rules behave when they are run on historical data in the order in which that data arrived. The study uses public datasets and publicly documented rules; no Rabobank data is required. A software workbench will implement the procedure, and a small data pipeline will show how its results can be used in practice.

2 Pilot work and change of research direction

The direction of this thesis follows from two pilot studies carried out in September and October 2026 [7] and a feasibility check carried out in November 2026 [8]. Both pilots used controlled error injection: values were changed on purpose in a copy of real data, and every change was recorded. Because the location of each error is known, it is possible to count which errors a control finds,

which it misses and how many alerts it raises on data that was not changed. The feasibility check instead ran rules on real reports without changing them. The figures below are internal pilot results, not independently published findings.

2.1 First pilot: choosing an error detector

The project began with a question about choosing error detectors for tables of data. Error detectors are methods that flag suspicious values without a separate rule being written for every possible error. For example, the CDC method predicts each value from the other columns of the same record and flags values that do not fit the prediction [9]. Earlier benchmark studies had shown large differences between such methods, and had shown that the effort needed to review their alerts is an important part of their cost [10, 11]. Jung et al. showed that the way errors are spread over the data can change how well a detector appears to perform [12].

The pilot therefore separated the type of change made to a value from the mechanism that decides which values are affected. Three mechanisms were tested. Erroneous completely at random (ECAR) means that every value has the same chance of being wrong. Erroneous at random (EAR) means that the chance depends on another recorded field, for example the source system that delivered the record. Erroneous not at random (ENAR) means that the chance depends on the value itself, for example when larger amounts are more likely to contain an error. The pilot applied missing values, invalid types and codes, amounts multiplied by ten and plausible substitutions to three public tables: credit-card clients, bank marketing and wine quality [13, 14, 15]. It compared rule-based and statistical controls with seven detectors, including CDC and Raha [9, 16], and measured how much of their added value remains when a reviewer can check only a limited number of alerts.

The error mechanism changed how many errors a detector found, but not which detector was best. The existing rules already found every missing value, invalid type and invalid code. On harder errors, the strongest additional detector found 11 to 21 of 40 changed values that the rules, the statistical checks and a simple prediction model missed, but it produced 520 to 900 extra alerts per batch of 800 records. With a budget of 20 reviews per 800 records, the extra alerts compete with the alerts of the existing controls for the reviewer's attention. The net gain, the number of extra errors found within those 20 reviews, was on average about 0.4 errors on the bank-marketing data, zero on the credit-card data and negative on the wine data. A negative gain means that the detector's alerts pushed useful alerts of the existing controls out of the review.

2.2 Why the research direction changed

The first pilot was not continued for four reasons. First, the mechanism changed the size of the effect but not the choice of detector, so there was little to select. Second, the question of what an extra detector adds when review time is limited had already been studied [10, 11]. Third, the public tables contain no record of real errors, so the pilot could only show that a detector recovers changes that we made ourselves. Fourth, the tables come without expert rules, so the pilot could not ask what an automatic method adds beyond the rules a real data team already has. The second pilot therefore moved to data that comes with an independent, published set of expert rules.

2.3 Second pilot: rules written by language models

The second pilot used financial reports that US listed companies file with the Securities and Exchange Commission (SEC). The Data Quality Committee (DQC) of XBRL US, an industry body, publishes validation rules for these reports together with their code and official test cases

[17]. Two small open-weight language models, Qwen2.5-Coder-7B and Qwen3-8B [18, 19], wrote rules in three ways: from the written description of a DQC rule, from an example report, and for types of error that the existing controls missed.

Three results stand out. First, reproducing the behaviour of the official rules was unreliable. For rules published before the models were released, the generated rules produced about 41% of the messages that the official test cases expect; for rules published afterwards, only 3% and 13% for the two models. Generating the same rule again often gave a very different result. Second, the generated rule sets were too noisy to use as a whole. The existing controls detected 27 of 83 injected errors; the best generated rule set detected six further errors, but raised on average 458 alerts per report that had never been corrected, against 1.8 for the DQC rules. Third, real corrections proved hard to use as ground truth. Of 20 numbers that companies had later corrected, only 9 were clearly wrong in the original report, and 7 of these were accounting restatements rather than data errors.

2.4 Follow-up: judging individual rules

A follow-up experiment looked at the causes and at individual rules instead of complete rule sets. When the model received the official code of nine rules it had failed to reproduce, it fixed three. Most of the remaining failures came from missing domain knowledge, such as information that is only available in the XBRL dictionary of reporting concepts, rather than from missing documentation. When individual generated rules were selected only if they detected errors that the existing controls missed, within a limit of one alert per report, one rule remained. It detected three additional injected errors at about 0.2 alerts per report. A generic filter that ignored the existing controls needed about twice as many alerts for the same gain. This is a small exploratory result and does not establish that the method is better. It does show that useful rules can be hidden inside noisy rule sets, and that selecting them relative to the existing controls is worth testing with stronger models, more data and a proper held-out evaluation.

2.5 Feasibility check: testing rules on historical data

The last step tested whether rules can be evaluated on historical data in the order in which it arrived, as a data pipeline would see it. Twenty-nine SEC reports filed between February 2023 and March 2025 were processed in filing order. The official DQC rules and the generated rules from the second pilot were run on each report, and every alert was linked to its report, its rule, the rule version and the date from which the rule existed. This worked for 98% of the alerts, and the official rules took about two minutes per report. Three results matter for the design. First, 46% of the DQC alerts came from rules that were published only after the report had been filed, so a test with today's rules overstates what a team could have caught at the time. Second, reports from early 2023 need an older release of the DQC rules. Third, none of the 13 checked corrections was flagged by any rule, expert or generated; most of them were accounting restatements, which validation rules are not designed to catch [8].

A separate short study tested whether a language model could predict which validation rules of the European Banking Authority would later be switched off as incorrect. It did not produce a reliable signal and was not selected as the thesis direction [7].

2.6 What the pilot work shows

Table 1 summarises how the pilot results shape this proposal. Together they suggest that there is something to gain: a small number of generated rules can add detections at a low alert cost. They also show that this gain only becomes visible when rules are judged one by one, relative to

the controls that already exist, and when they are tested in a way that matches how they would be used.

Pilot result	Consequence for this thesis
Simple rules already caught all format and code errors; extra detectors mainly added alerts.	Every method is compared with the existing controls, and alerts are always reported next to detections (RQ2).
Generated rules reproduced older official rules much better than newer ones.	Rules are split by publication date relative to the model's release date (RQ1).
Most failures came from missing domain knowledge, not missing documentation.	One experimental condition adds domain knowledge to the model's input (RQ1).
Whole generated rule sets were too noisy, but one selected rule added detections at 0.2 alerts per report.	Selecting rules relative to the existing controls is the main proposed method (RQ2).
Most real corrections were accounting restatements, and none of the 13 checked corrections was flagged.	A larger, carefully reviewed set of real errors is built, weighted towards likely data errors (RQ3).
Almost half of today's alerts came from rules that did not yet exist at filing time.	The historical test distinguishes rules available at the time from rules developed later (RQ3).

Table 1: Pilot results and their consequences for the proposed research.

3 Related research and proposed contribution

This section describes five lines of earlier work. For each line it explains what the studies do, how they judge their results, and what they leave open for this thesis. Table 2 at the end of the section compares the most relevant studies.

3.1 Detecting errors and the cost of reviewing alerts

Abedjan et al. ran several error detection tools on real datasets in which the errors were known. Each tool found only part of the errors, and combining tools found more errors but also produced more alerts that a person had to verify [10]. REIN later compared a large number of detection and cleaning methods on the same datasets [11]. The study by Abedjan et al. in particular shows that a new check should be judged by what it adds to existing checks and by the review effort it creates, not by its detections alone. Both studies evaluate data-driven detectors, however, not rules written by language models.

3.2 Language models that write data-quality rules

Alexander et al. asked language models to write data validation tests, small programs that check a dataset, and compared them with a test suite written by one data scientist. They repeated each setup five times to measure how consistent the generated tests were [20]. Akella et al. combine statistical screening of the data with retrieved domain information: the model proposes rules, translates them into executable code, and presents each rule on a readable card for review [21].

LeDQeR asks a model to generate candidate rules for Great Expectations, an open-source tool in which checks are written as "expectations" about a table, starting from an example of an erroneous record. Four filters then remove rules that do not run, that are wrong on labelled data, that only fit a single record, or that repeat another rule. The rules are scored on benchmark tables

in which every error is labelled [22]. AuDQ uses several cooperating "agents", language-model programs that plan steps, call tools and check their own output, to discover rules. The authors also describe how rules can be versioned and traced to their origin, and they let people rate the rules. They note that their datasets have no reference rule set to compare against [23].

Two master's theses address the same task. Schneider guides the model with templates that fix the shape of a rule and compares the output with five to ten expert "golden rules" per dataset, judging by hand whether each golden rule is covered [24]. Siyam compares several commercial models and prompting strategies for generating data-quality rules [25].

These studies establish that rule generation, filtering and human review with language models are existing research directions. They judge the generated rules against labelled benchmark errors, human ratings or a small set of reference rules. None of them compares generated rules with a large, independently maintained rule set using that rule set's own executable tests, and none asks what a generated rule adds beyond the controls that are already running.

3.3 Selecting a small set of useful checks

Generating many candidate checks is easy; choosing the few that are worth keeping is harder. SPADE addresses this for applications built on language models. It derives many automatic checks on the model's output from the changes developers made to their prompts, and then selects a small subset that still catches the failures developers have labelled, while keeping the share of false failures below a chosen limit [26]. Auto-Validate learns patterns for text columns, such as the format of a date or a product code, from a large collection of tables, and chooses patterns that keep the expected number of false alarms low [27]. LeDQeR's last filter removes a generated rule when another generated rule already covers it [22].

These methods judge candidates on their own or against each other. None of them selects relative to an existing set of expert controls, so they may keep checks that only repeat what those controls already catch. This is the gap that the main method of this thesis addresses.

3.4 Checking recurring data and testing on history

Many data sources deliver new data in recurring batches. Redyuk et al. learn from statistics of earlier batches what a normal batch looks like and flag new batches that deviate [28]. Auto-Validate-by-History, developed at Microsoft, uses the history of earlier runs of a recurring pipeline to choose quality constraints with a target false-alarm rate, and was evaluated on about 2,000 production pipelines [29]. Both use history to build and test statistical checks; neither tests rules written by language models against expert rules.

Modern data platforms, often called lakehouses, store data as files and add database features such as versioned tables [30, 31]. In such platforms, new data can first be written to a separate area, checked, and only then published. Tagliabue and Greco argue that AI agents should change a data platform only through such isolated and checked steps [32]. Benchmarks show why care is needed: in ELT-Bench, the best AI agent at the time of publication built only 11.3% of the required data models correctly [33]. In banking, testing a model on past data before trusting it, known as backtesting, has long been standard practice for risk models [34]. We did not find work that backtests generated data-quality rules in this way, or that uses such a test to decide whether a rule may stop data.

3.5 Evaluating code written by language models

Three effects complicate the evaluation of generated rules. Language models perform worse on specialised programming languages with little public code, such as the rule language used for financial reports [35]. Their output can differ between repeated runs [5]. And public rules may have been part of their training data, so a test on such a rule may measure memory rather than skill; benchmarks for generated code therefore separate test items by publication date [36]. For the OLMo models, whose training data is public, it is also possible to search the training data directly for a given text [37, 38].

3.6 Proposed contribution

Table 2 summarises the comparison. Systems that generate data-quality rules with language models also exist commercially and have been patented [4, 39]; the contribution of this thesis is therefore not the idea of such a system. The anticipated contribution is an empirical evaluation with four parts: (1) generated rules are scored against independently maintained expert rules, using those rules' own tests and controlling for rules the model may have seen; (2) a method that selects generated rules by what they add beyond the existing expert controls is compared with alternative selection methods at fixed alert budgets; (3) the selected rules are tested on historical data in the order in which it arrived, using reviewed real corrections; and (4) a reviewed set of real errors in financial reports is produced. This claim is provisional and will be reassessed against literature published during the thesis.

Study	What it does	How results are judged	Accounts for existing expert controls	Tested on history
Alexander et al. [20]	Generates data tests	One suite by one data scientist; five repeats	No	No
Akella et al. [21]	Generates rules and code with retrieved knowledge	Benchmark data; rule cards	No	No
LeDQeR [22]	Generates and filters rules	Labelled benchmark errors	No	No
AuDQ [23]	Agents discover rules; provenance	Human ratings	No	No
Schneider [24]	Template-guided generation	5 to 10 expert rules per dataset, judged by hand	No	No
SPADE [26]	Selects checks on LLM output	Labelled failures; false-failure limit	No	No
Auto-Validate-by-History [29]	Learns statistical constraints	Past pipeline runs	No	Yes
This proposal	Generates, selects and backtests rules	Official expert tests; injected errors; reviewed real corrections	Yes	Yes

Table 2: Comparison with the most relevant earlier work, as far as reported in the cited sources.

4 Research questions

Main research question: *Under what conditions can validation rules generated by open-weight language models reproduce expert-defined checks, provide additional error detection beyond existing expert controls within a practical alert budget, and be used safely in a data pipeline?*

RQ1. How accurately and consistently can language models implement documented expert rules,

and what is the effect of adding domain knowledge and of letting the model test its rule on development examples?

RQ2. How much additional error detection do generated rules provide beyond the existing controls, how do five selection methods compare at fixed alert budgets, and can the evidence from this selection decide whether a rule should block data, only warn, or only be monitored?

RQ3. When the selected rules are run on historical financial reports in the order in which they were filed, how many reviewed real errors do they detect, how many alerts do they raise, how stable are they over time, and does the chosen level of action still hold on later data?

RQ1 tests whether a model can reproduce what experts already wrote; it follows from the reproduction and failure results of the second pilot. RQ2 tests whether generated rules can add something that the experts did not write; it follows from the follow-up experiment, in which one selected rule added detections at a low alert cost. RQ3 tests whether those results still hold on real data over time; it follows from the feasibility check.

5 Data sources and reference material

5.1 What "independent expert rules" means

Each public dataset in this study comes with validation rules that were written and published by an organisation outside this project. These organisations maintain the rules for real reporting, document them, and in most cases also publish code or test cases that show what each rule should report (Table 3). Because neither the author nor the language models wrote these rules, they can serve as an independent reference: a generated rule can be compared with them, and they form the existing controls against which any additional value is measured.

Organisation	Who they are	What they publish
XBRL US Data Quality Committee (DQC)	Industry committee for US digital financial reporting	About 200 rules for SEC reports, with code and official test cases [17]
XBRL International	International standards body for XBRL	Data-quality rules for European company reports, with test cases and expected results [43, 44]
FFIEC and CFPB	US financial regulatory bodies	"Edits" for mortgage reporting, with open-source validation software and tests [45, 46]
GLEIF	Operator of the global register of legal entities	207 written data-quality checks, without code [47]

Table 3: Organisations that wrote the expert rules used in this study.

5.2 Datasets

D1: US company reports and DQC rules. The SEC is the US regulator that receives the reports of listed companies. These reports are filed in XBRL, the eXtensible Business Reporting Language. In XBRL every number is a separate fact with a standard concept name, a reporting date or period, a unit and a level of rounding, for example the concept `Assets` with a value of 340,000,000 US dollars on 31 December 2014. A comparison between facts must therefore respect their accounting meaning, not only their values.

Rule DQC_0004, for example, checks that assets equal liabilities plus equity, and its documentation gives the message "Assets with a value of 340,000,000 is not equal to the total of Liabilities and Equity with a value of 350,000,000" [17]. The DQC rules are written in the rule language XULE and run with the open-source XBRL processor Arelle [17, 40]. Of these rules, 153 have official test cases; 38 were first released after 29 April 2025, the release date of Qwen3, which makes it possible to test rules that a model released before that date cannot have seen. The SEC Financial Statement Data Sets show which reports were later amended: from 2023 to mid-2025, 1,294 of 68,875 reports were followed by a corrected version [41]. D1 is the main dataset, because it has the most complete reference and the pilot software already runs on it.

D2: European company reports and the ESEF rules. ESEF, the European Single Electronic Format, is the format in which issuers listed on EU regulated markets publish their annual financial reports [42]. Consolidated statements prepared under International Financial Reporting Standards (IFRS) are tagged in this format with Inline XBRL. XBRL International publishes data-quality rules for these reports that are derived from the DQC rules [43]. Of the 25 rules in the current set, 22 have official test reports, and an official file lists the expected results of the rules on 1,739 real European reports from 2022 [44]. D2 tests whether results from the US setting also hold under different accounting standards and for different companies.

D3: US mortgage applications and HMDA edits. Under the Home Mortgage Disclosure Act (HMDA), US lenders above certain size thresholds report data on the mortgage applications they receive. The regulators publish "edits", validation rules described in plain language, of four kinds: syntax edits check the file format, validity edits check that fields contain allowed values, quality edits flag unusual values that the lender must confirm, and macro quality edits look at a file as a whole [45]. Quality edit Q604, for example, checks that the reported county exists in the census list for the reporting year. The regulator's own validation software is open source and contains 318 edit codes and 330 test files [46]. The public loan data has been modified for privacy and has already passed the filing controls, so only suitable fields and controlled error injection will be used. D3 tests a source that is closer to the ordinary tables in a bank's data platform.

D4: The GLEIF register of legal entities. The Global Legal Entity Identifier Foundation (GLEIF) maintains the register of Legal Entity Identifiers (LEIs). An LEI identifies a legal entity, such as a company, across countries, and banks use it to identify the parties they deal with. GLEIF publishes 207 data-quality checks, each written as a precondition, a condition and an expected outcome; a check can pass, fail or not apply [47]. Check C000020, for example, fails when a legal name is "n/a" or consists only of punctuation. GLEIF does not publish code or a result for each record, so a small test suite must be built (Section 5.4). D4 is optional if time is limited.

D5: Reviewed real errors in SEC reports. D5 will be built from SEC reports that were later corrected. A changed number is not necessarily an error that could have been detected in the original report; it may reflect a later accounting restatement or other new information. D5 is therefore planned as a reviewed resource, not as a verified gold standard from the start (Section 5.4).

5.3 What counts as the correct answer

To score a rule, a reference is needed that states the correct answer, often called the ground truth. The three research questions need different references (Table 4). For RQ1 the expert rule is the reference: a generated rule is correct when it produces the same messages as the official rule on the official test cases. For RQ2 and RQ3 the expert rules cannot be the reference, because they also miss errors; in the second pilot, all existing controls together detected 27 of 83 injected errors. For these questions it is necessary to know where errors in the data really are.

Two limits of real corrections shape the design of RQ3. First, a report that was never corrected is not necessarily free of errors, so an alert on such a report is not automatically a false alarm.

Question	Reference	Main limitation
RQ1: does the generated rule behave like the expert rule?	Official test cases (D1 to D3); constructed test records (D4)	Shows what the rule should report, not whether the data is wrong
RQ2: what does the generated rule add?	Injected errors with known locations	Exact, but the errors are not real
RQ3: what would the rule have done on real data?	Reviewed corrections (D5) and a reviewed sample of alerts	Small; a report that was never corrected is not proven correct

Table 4: Reference used for each research question.

Alerts on these reports are therefore treated as a measure of review burden, and the true share of false alarms is estimated from a reviewed sample. Second, a correction does not prove that the error was visible in the original report. In a restatement the company revises its accounts, so the original number matched the accounts as they stood at the time. Each reviewed item therefore also records whether the issue could reasonably have been detected with the information in the original report.

5.4 Reference material built during the project

Reviewed real errors (D5). The target is 300 corrected numbers, spread over types of correction and over companies, with extra weight on types that are more often data errors, such as a reversed sign, a factor of 1,000 or a shifted reporting period. For each number, a reviewer opens the original and the corrected report and records whether the original number was wrong, the type of correction (tagging or period error, typing error, accounting restatement, not an error, or unclear) and whether the issue was visible in the original report. The author reviews all 300 items without labels suggested by a language model, following a written guide with examples. A second reviewer, preferably a professional with accounting or XBRL experience, reviews the same 100 items independently. Agreement is measured with Cohen's kappa, a standard measure that corrects for agreement by chance [48], and disagreements are reported. If no independent reviewer is available, the study will report the lack of an external gold standard and will not present the author's labels as verified fact.

Test records for GLEIF (D4). For 40 to 60 checks, three small test records are written per check from the published specification: one that should pass, one that should fail and one to which the check does not apply. A second person reviews them.

Reviewed alert sample (RQ3). From the alerts raised on reports that were never corrected, a random sample of 200 is drawn and reviewed with the same guide, to estimate how many of these alerts point to real problems.

6 Experimental design

6.1 RQ1: Reproducing expert rules

For each rule with a reference, a model will write an executable implementation under three input conditions: (A) the written description of the rule and the necessary data structure; (B) the same information plus relevant domain knowledge, such as whether an amount is normally positive, whether it describes a moment or a period, and which amounts add up to which total; and (C) condition B with permission to run the rule on development examples and repair it. The official

test outcomes will never be given to the model. Each task will be repeated five times with fixed generation settings to measure variation. The generated messages will be compared with those of the official implementation, reporting the share of expected messages found (recall), the share of produced messages that are correct (precision), exact agreement per test case, the share of code that does not run, and the variation between repeats.

Rules will be split by publication date relative to the release date of each model. This reduces the risk that results reflect memorisation of public rules, although it cannot exclude related knowledge learned earlier. For OLMo, whose training data is public, the training data will also be searched directly for the text of each rule [38]. In addition, a language model will judge, by reading only, whether each generated rule is equivalent to the expert rule, as in earlier work that judged coverage by reading [24]. Comparing this judgement with the execution results shows how often a rule that looks correct does not behave correctly.

6.2 RQ2: Additional detection and rule selection

The baseline consists of the official expert rules, basic format checks and inexpensive statistical checks learned from other reports, for example for unusual signs or sudden jumps between periods. In a copy of the data, errors will be injected such as sign changes, magnitude changes, altered digits, wrong concepts or codes, swapped reporting periods and omitted values. Each injected change is recorded at the level of the affected fact or field, and where real corrections are available they inform the size of the injected changes. Results on injected errors will be kept separate from conclusions about real error rates.

Models will then generate candidate rules for the error types that the baseline misses. Candidates are generated on development material and evaluated on unseen companies. Five selection methods will be compared using the same candidate pool, the same observations and the same budget: (1) keep every rule that runs; (2) a generic filter based on validity and alert volume, in the style of LeDQeR [22]; (3) a compact selection that balances coverage and false alarms, inspired by SPADE [26]; (4) a language model that rates the plausibility of each rule; and (5) an expert-aware selection that keeps only rules detecting errors the baseline misses, while also considering alert volume and repeatability. The fifth method is the main proposed method; its scoring and thresholds will be registered before the final evaluation.

The primary outcome is the number of additional detections at fixed average alert budgets of 0.5, 1 and 2 alerts per report. The budget makes the practical trade-off explicit: a rule that detects ten extra errors but raises 450 alerts per report is of no use to a team that can review one alert per report. Secondary outcomes include recall, precision where clean labels exist, alerts on reports that were never corrected, runtime, the share of generated rules that fail, and stability across repeats. From the same evidence, each selected rule receives a proposed level of action: block (hold the data), warn (let the data pass with an alert) or monitor (record only). The thresholds for these levels are fixed in advance; whether the levels are reliable is tested in RQ3, and a person always takes the final decision.

6.3 RQ3: Testing selected rules on historical data

A sample of about 2,000 SEC reports will be processed in the order in which they were filed, in weekly batches. For each batch, the official rules and the selected generated rules are run and every alert is stored with its report, rule, rule version and availability date. Based on the feasibility check, this takes about two minutes per report for the official rules, or roughly 70 hours of computing time in total. Two forms of historical test will be distinguished, because they answer different questions:

1. A historical stress test applies rules developed now to past reports. It shows how a rule behaves on real data from different periods.
2. A time-bound backtest uses only what was available at the time: the DQC release that was current on each filing date, generated rules selected on earlier batches only, and a model released before the period being replayed. It will therefore start in May 2025, after the release of Qwen3. All DQC releases are public [17]; if older releases do not run with the current software, the current rule code will be filtered by release date and this will be reported as a limitation.

The following configurations will be compared: the expert rules alone; the expert rules plus all generated rules; and the expert rules plus the rules kept by each of the five selection methods. The outcomes are the reviewed real errors from D5 that would have been flagged in the original report, the number of alerts per week on reports that were never corrected, the share of false alarms estimated from the reviewed sample, stability across quarters, and whether a level of action chosen on earlier data still holds on later data. Recent reports have had less time to be corrected, so results will be reported per period.

6.4 Professional review of the evidence

Three to five data-quality or accounting professionals will review a limited set of evidence cards produced by the workbench (Section 8). They will record whether they would accept, reject or revise each rule, at which level of action, and how long the decision took. Their decisions will be compared with the automatic proposals. Reviewer recruitment, ethics approval from the faculty and permission to release derived labels will be arranged before this stage. This part is an extension if reviewers are not available in time.

6.5 Protocol, data splits and statistics

The data splits, alert budgets, outcome measures and selection thresholds will be written down in a dated protocol before the final evaluation. Development and test material will be separated by company, and by time where relevant, so that rules are never selected and evaluated on reports of the same company. Results will be reported with 95% confidence intervals obtained by resampling over companies or rules, with paired comparisons where methods run on the same data. All runs will be reported, including failed runs and results that go against the expectations.

7 Models, baselines and reproducibility

The project will use open-weight language models that can be run on controlled infrastructure (Table 5). Such models can be downloaded and run locally, so no data has to be sent to an external service, the exact model version can be stored, and the release date is known, which the contamination analysis requires. The larger experiments will run on the university's computing cluster Hábrók, where software runs in Apptainer containers [49]; the small models remain usable on an ordinary computer. Self-hosting avoids a mandatory external inference service, although computing and maintenance still have costs.

Model versions, generation settings and prompts will be fixed and documented. All selection methods see the same generated candidate rules and the same development labels; only the use of the existing expert controls differs. This prevents a stronger generator or extra test knowledge from being mistaken for a better selection method. As a secondary baseline, a language model will also inspect reports directly and point out suspicious facts, without producing reusable rules.

Model	Released	Role
Qwen2.5-Coder 7B and 32B [18]	2024	Specialised in code; the 7B version provides continuity with the pilots.
Qwen3 8B and 32B [19]	April 2025	General model; the 8B version provides continuity with the pilots.
gpt-oss 20B, optionally 120B [50]	August 2025	Second model family, to test whether results depend on one developer.
OLMo 2 7B and 32B [37]	2024–2025	Public training data, which allows a direct contamination check.

Table 5: Planned models. "B" stands for billions of parameters, a rough measure of model size.

This makes the difference between one-time model judgements and reusable, executable checks explicit.

8 Engineering artefact and organisational use

The engineering deliverable is a reproducible workbench built from the pilot software, together with a small data pipeline that uses its results (Figure 1). A user registers a data source and its existing checks, runs a stress test on a safe copy, lets a model generate candidate rules, executes them in a restricted environment, and receives evidence for each rule. The evidence includes the rule text, the information used to generate it, the execution status, the additional detections, the alert burden in the historical test, the stability across repeats and across periods, and the proposed level of action. A reviewer then accepts, rejects or requests changes, and every decision is stored in an append-only event log. The log is kept separately from the data tables, because older table versions are removed after a retention period, by default seven days [52].

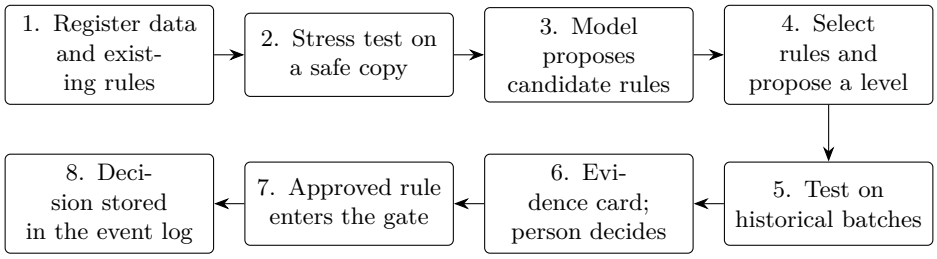


Figure 1: The workbench. It never adopts a rule by itself and never changes data.

The pipeline demonstrates how approved rules are used. It is built with Apache Spark and Delta Lake, two open-source technologies that are widely used in data platforms [51, 31]. New SEC reports arrive in a staging area, the official rules run through Arelle, the approved generated rules run next, and each batch is then published, published with warnings, or held back for review, according to the levels of the rules that fired. The workbench uses adapters for the rule systems in this study and for ordinary tables, and an interface to replace one approved local model with another. Generated code runs without production credentials, and the tool does not modify data or deploy rules automatically.

For Avanade and Rabobank, the intended use is the evaluation of proposed controls when a data source is onboarded or maintained. A team can see which kinds of error its existing controls miss, review a small number of tested candidate rules, and test any new rule, generated or hand-written, on its own historical data before it goes live. The software would still require internal security approval, data-specific adapters and integration work before production use. The project does

not assume access to confidential banking data, and it does not promise measurable operational savings before user testing.

9 Scope, feasibility and expected outputs

The project runs for nine months (Table 6). At the end of month 3, progress will be reviewed. The minimum scientific scope is the SEC/DQC evaluation for all three research questions, at least two model families, a controlled comparison of the selection methods, and an independently reviewed subset of real errors. HMDA is the preferred second domain, ESEF a related financial-reporting replication, and GLEIF an extension if time and reviewer capacity permit. If the programme becomes too large, GLEIF, then HMDA, then the depth of the ESEF analysis, and then the number of models will be reduced before the held-out evaluation is compromised.

Months	Work
1	Protocol, literature update, ethics request, cluster access, review guide, rule register and historical test engine
2–3	RQ1 and RQ2 on SEC/DQC; GLEIF test records; start of the D5 reviews; progress review at the end of month 3
4	RQ1 and RQ2 on HMDA and ESEF; data pipeline with the quality gate
5–6	RQ3 historical tests; completion of D5 and the reviewed alert sample; evidence cards and professional review
7	Analysis across datasets; robustness checks; demonstration to Avanade and Rabobank
8–9	Writing, documentation and contingency

Table 6: Nine-month plan.

Risk	Response
Generated rules add little, even with larger models	The thesis is a measurement; a clear negative result with its causes is informative.
Few reviewed real errors were visible in the original report	Weight D5 towards correction types that are more often data errors; report RQ3 with uncertainty intervals.
Reports that were never corrected are treated as correct	Use their alerts as a measure of review burden only; estimate false alarms from the reviewed sample.
The historical test uses information from the future	Store availability dates for every rule and model; run the time-bound form only after the model's release.
Building the pipeline takes time from the research	Keep the pipeline small and limited to SEC data; the scientific results come from RQ1 to RQ3.
Reviewers are not available in time	Request ethics approval in month one; report the limitation if no independent review is possible.

Table 7: Main risks and responses.

The expected outputs are a thesis that reports both successful and unsuccessful results, a reproducible experimental benchmark and selection procedure, a reviewed set of real errors subject to permissions, and a tested workbench and pipeline with documentation. An outcome in which generated rules do not provide reliable additional detections remains informative, provided that the controls, test data and limitations are reported transparently.

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